

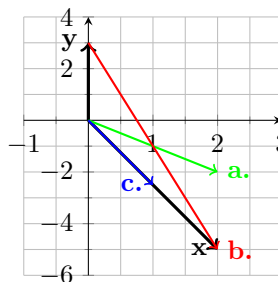
1. Let $\mathbf{x} = (2, -5)$ and $\mathbf{y} = (0, 3)$. Complete the following:

a. $\mathbf{x} + \mathbf{y} = (2 + 0, -5 + 3) = (2, -2)$

b. Sketch $\mathbf{x} + \mathbf{y}$.

c. Sketch $\mathbf{x} - \mathbf{y}$. $(0, -8)$

d. Sketch $\frac{1}{2}\mathbf{x}$. $(1, -5/2)$



2. Determine if the point $(-6, 2)$ is on the line defined parametrically by $\mathbf{x} = (0, 1) + t(-3, 1)$.

$$(0, 1) + t(-3, 1) = (-6, 2) \Rightarrow 0 + -3t = -6 \Rightarrow t = 2.$$

So then $(0, 1) + 2(-3, 1) = (-6, 3) \neq (-6, 2)$. So the point is *not* on the line.

3. Let $(1, 1, 1)$ and $(2, 0, 1)$ be two points on a plane H . Find a line on the plane.

$(1, 1, 1) - (2, 0, 1) = (-1, 1, 0)$ is the direction vector of the line connecting the two points. So then $\ell = (1, 1, 1) + t(-1, 1, 0)$ for $t \in \mathbb{R}$ is on the plane (answers may vary).

4. Let $\mathbf{x}$ and $\mathbf{y}$ be two vectors such that $\mathbf{x} \cdot \mathbf{y} = 0$.

a. What is the angle between $\mathbf{x}$ and $\mathbf{y}$?

The dot product is 0 meaning that the vectors are orthogonal. So the angle is $\frac{\pi}{2}$ or 90° .

b. If $\mathbf{x} = (1, 1)$ find a vector parallel to $\mathbf{y}$.

$(1, 1) \cdot (y_1, y_2) = 1y_1 + 1y_2 = 0 \Rightarrow y_1 = -y_2$. So $(1, -1)$ is parallel to $\mathbf{y}$ (answers may vary).

Note that and vector parallel to $\mathbf{y}$ will be equal to $c\mathbf{y}$ for some scalar $c \in \mathbb{R}$.